

- CHAPTER 12 -

MESOCYCLE PLANNING

INTRA-MESOCYCLE FACTORS

Intra-mesocycle factors deal with structuring weekly workouts in terms of your ability level. The progression charts in this book specify four distinct levels of ability. This chapter addresses the factors you need to consider at each level.

- Beginner Levels: 1-5
- Intermediate Levels: 6-9
- Advanced Levels: 10-13
- Elite Levels: 14-16

A typical mesocycle restructure may look like this:

	Microcycle – 1 Week
	Microcycle – 1 Week
Mesocycle	Microcycle – 1 Week
	Microcycle – 1 Week
	Microcycle – 1 Week
	Rest/Deload Week

Weekly scheduling is generally the preference since we are conditioned to think in terms of weeks for most things in life. However, you can run a routine on a five-day, six-day, or two-week schedule. It's up to you.

Workout routines are constructed over the course of four to eight weeks, followed by a deload period. Learning how to structure your routine in these periods is going to be critical to avoiding injuries. In addition, it will help you adjust to how your body responds to training in the context of plateauing. You will have to learn to utilize different types of programming as you progress. If you are keeping a training log (which you should be doing), you will be able to figure out what works best for you in terms of programming. This will give you a clearer understanding of how to program your own training, avoid pitfalls along the way, and learn how to be a better coach—even if you are only coaching yourself.

Programming on the elite level is not addressed in this chapter. Once you are that strong, you should know how to apply programming concepts with upper-level strength skills. At that level, strength consists

more of working to overcome specific weaknesses and working combinations or sequences of strength skills. Transitional movements become important as moving in and out of strength skills requires phenomenal body control and works your muscles in ways that they are not usually worked.

BEGINNER LEVEL: EXERCISES AND RECOMMENDATIONS

Most people wanting to learn bodyweight exercises will start in the beginner ability level. The beginner level (levels one through five) covers most of the basic bodyweight exercises from basic wall handstands, pushups, dips, and handstand pushups, to pull-ups, ring rowing, muscle-ups, and some of the basic progressions of isometric holds.

Even if you have an athletic background or have pursued other strength and conditioning endeavors before pursuing bodyweight training, you may fall into this category. It is likely that you still need to train in some of the skills such as handstands and the isometric positions even if you already have the requisite strength to start above this range. These basic skills require lots of practice, and they slowly build up your connective tissues for the exercise progressions themselves. If you are coming in with a decent amount of strength, focus your efforts on mobility work and joint preparation.

Eccentric exercises are not recommended at the beginner level, as they can be tough on joints and connective tissues. There are two exceptions: using eccentric exercises to help gain the strength for pull-ups or dips. There are many different types of simple progressions for concentric and isometric exercises that are better for a beginner's body and strength than eccentrics.

The lowest level of progressions (levels one through three) are best for women, who tend to have a lower strength to bodyweight ratio starting out, and for anyone currently on a weight-loss program. Everyone starts somewhere! Learning and accomplishing these exercises will build a strong foundation, which can be built upon as you reach higher ability levels. You will be surprised at what you will be able to accomplish when you give proper focus to improving and refining your workouts.

After these two big groups, joint and connective tissue preparation and mobility will be the main factors that must be taken into account. Aside from focusing on linear progression or linear repetition progression, flexibility and mobility should be undertaken. As you look at the charts you will notice that the L-sit progression requires an increasing degree of compression (hamstrings/hip flexibility) to move from the L-sit to the V-sit progressions. Other skill-based strength movements will also be critical to the development of strength (such as straight-arm press handstands) and will also require significant amounts of compression. To improve mobility, you should aim to get your chest to your knees in the pike and straddle positions, and your hands to the floor when standing and bending over.

Higher repetitions are initially good for untrained beginners. In the first few months of exercising regularly, it may be a good idea to build up to 15-20 repetitions in your workouts before moving on to the next progression. This ensures that you get enough practice to establish good techniques. The high repetition count will also help keep your connective tissues healthy.

You should focus your mobility and strength work on a few key areas: your shoulders, elbows, wrists, and thoracic spine. Refer to Chapter 11 (where we discussed prehabilitation) if you need advice on how to program these into a routine. Refer to Part 5 for the list of mobility and flexibility exercises.

BEGINNER LEVEL: WEEKLY SCHEDULING

For all beginners (levels one through five), the simple intra- and inter-exercise progressions should be used in order to progress from one workout to the next. There is no better time than as a beginner to use progressions to improve your abilities from workout to workout. Leave the more complex concepts for later. You should begin with linear progressions or linear repetition progressions. These will take you far. Once you begin to plateau with these progressions, you can select a different option. If you begin to regress in your ability level and become lethargic, you may need a deload period to recover from fatigue.

In general, you can expect to progress with most of the exercises every workout. The main thing you need to watch out for is connective tissue integrity. If you are having difficulty, you may want to scale back a bit with your progressions and focus on mobility and prehabilitation work instead. Remember, physical health is more important than progress! If you are unhealthy, you will not be able to make much progress. Rapid progression should never come at the expense of your health.

As a beginner, the most important key is *consistency*. A lack of consistency would be defined as performing workouts only once or twice a week, or skipping weeks entirely. If you are sporadic in your training, you will likely not progress. It is common to get stuck here for five years or more, but this does not have to happen to you! Starting a routine and being consistent with it, in whatever state, is better than waiting until your routine or circumstances are perfect. Do not allow yourself to get caught up with minor details that may make your programming better, as programming is not complex at this stage. From cycle to cycle, your exercises will not change much. If you start over-thinking everything it can lead to paralysis by analysis.

Jump right in to a full-body program with three workouts per week. This gives you a day or rest and recovery between workouts, with two off days after every three workouts. This structure will typically operate on a M/W/F or Tue/Thur/Sat schedule, but other variations can be used as well.

When constructing your initial routine, start with your goals, then follow the recommendations of the book, be sure to take into consideration which exercises you will do and what your weekly schedule is outside of your workouts.

The needs of a *rank or untrained beginner* are simple:

- Learn the fundamental exercises and attain a good proficiency in them.
- In most cases, utilize a higher number of repetitions to solidify movement patterns and build up connective tissue strength.
- Focus heavily on the personal weaknesses you bring to your workouts. For example, if you have a desk job, you may have issues stemming from poor posture that could lead to injury if left unaddressed. Most sedentary individuals tend to have very poor mobility and flexibility.
- Put yourself on a generalized, balanced routine that will begin with high repetitions and then transition to traditional strength training.

The needs of a *trained beginner* are simple:

- Emphasize consistency in training. Discipline is going to be the most important factor in making progress. It is not good to skip workouts, apart from potential overuse injuries or emergencies. There is some merit to the common saying: *The best program is the one you stick to.*

- Emphasize training that ensures good muscular development and strength, especially in the 5-8 repetition range
- Make sure there is a proper balance of both pushing and pulling exercises in your routine.
- Add in exercises to maintain structural balance if imbalances start to develop. Typically, this will be adding more horizontal pulling if you trained mostly pushing exercises before coming to body-weight training.
- Allow your body to accommodate to strength training to allow your connective tissues and underlying structures (such as your joints and bones) to adapt.

BEGINNER LEVEL: ENDING A CYCLE

There are two equally valid ways to terminate a mesocycle. Since linear progression and other simple methods can often be continued for many consecutive months, one scenario is to end a mesocycle when you hit a plateau phase in your training. In terms of scheduling, that would be after 1 Week has gone by without progression or increase in repetitions for all exercises. The alternative is to end the cycle after four to eight weeks, which allows a rest week for recovery.

This second scenario (ending after 4-8 weeks) is preferred if your joints or connective tissues are starting to get sore or overused. It is never a good idea to train your body into oblivion. In the beginning, your joints and connective tissues can be the limiting factor rather than actual nervous system fatigue or muscle recovery. However, if you can continue making progress, continue with your cycle.

For the most part, staying on four to eight-week block schedules tends to be a good framework. This schedule allows you to evaluate your progress, make any necessary changes, and give your body a break to deload. Do not be afraid to slow down your training. It is not a race but a journey.

INTERMEDIATE LEVEL: EXERCISES AND RECOMMENDATIONS

The intermediate ability level (levels six through nine) develops the strength and body awareness to execute some of the commonly desired gymnastics progressions. Namely, we are working on solid handstands and progressing toward one-arms, rings handstand work, freestanding handstand pushups, straight-arm press handstands, full back and front levers, straddle planche, slightly past vertical V-sit, and lots of other cool multi-plane pulling/pressing strength exercises.

The full range of motion concentrics and the isometrics are still at your disposal. At this point, the eccentric exercises tend to be useful for gaining strength, since pulling exercises tend to respond particularly well to them. Eccentrics can be particularly useful for the development of the back lever, front lever, one-arm chin-up, and other pulling variations. A pushing exercise they are also used for is straight-arm handstand eccentrics.

By this point you have worked your flexibility and mobility drills until your compression (hamstring and hips) is becoming very good—approaching maximal levels. Your chest can touch your knees in pike and to the ground in the straddle position, and your hands can be flat on the floor while standing with your legs

straight. If this is not the case, focus on these abilities until you can accomplish them. You will be unable to progress past intermediate level without these skills.

Once you have achieved this level of flexibility/mobility, preparing your joints and connective tissues plays a greater role since you will soon be moving on to strength-based moves that are harder on your joints, namely, the iron cross, one-arm chin-up, back lever, and planche. These exercises can lead to serious injuries of the shoulders, elbows, or other parts of the body if you do not adequately prepare by developing the specific muscular strength needed and conditioning your connective tissues adequately.

On the charts, the progressions highlighted in gray will help you increase the strength of the connective tissues in your shoulders and elbows in the context of iron cross training. By following these progressions, you should reach a place of adequate preparation to begin attempting both the iron cross and one-arm chin-up. These progressions can also be used to gradually prepare for other types of movement that can be hard on connective tissues.

INTERMEDIATE LEVEL: WEEKLY SCHEDULING

By the intermediate ability level, you should have developed a solid base of both strength and conditioning. Typically, those who enter this level will have approximately 12-24 months of bodyweight training under their belt or more. For athletes with greater body mass (175 pounds or more), it may take longer to reach this level. Athletes with a strength sport background or a very solid barbell or weights background also should not rush to reach the intermediate level. These athletes must take it slow at the beginner level with the straight-arm exercises because, though they may easily have the raw strength, they do not yet have adequate connective tissue strength to perform straight-arm exercises at intermediate level. A lack of connective tissue strength makes you particularly susceptible to injury.

At the intermediate level, you can decide whether or not you want to add an extra workout to your weekly schedule. The general strategy for adding an extra workout is to add one for each year you have spent consistently doing strength and conditioning work. This is a safe and conservative recommendation. Certainly, extra workouts can be added earlier, but you run the risk of overtraining and burning out. If an extra day of workouts is added, your schedule may look like M/Tue/Thur/F or Tue/W/Sat/Sun. The other alternative is to add an additional exercise for each of the push and pull categories to your existing M/W/F schedule. If you are transitioning to a 4x per week schedule instead of 3x per week, drop down to two exercises per push and pull category for each workout for at least one cycle. Build up volume as you get stronger and your conditioning increases.

If you are adding an extra workout to your weekly schedule, make sure to track how you feel for the first few weeks. Look for signs of overtraining: loss of appetite, deterioration of sleep quality, decrease in motivation, and so on. If your body indicates that it is being overworked, it is a good idea to drop back down to a 3x per week schedule until you build a better strength and conditioning base.

Alternatively, you may be interested in switching to a split routine at this point to ensure that your connective tissues and other body parts get more rest. Such routines could include push/pull, upper/lower, or straight-arm/bent-arm. Three-part splits—like push/pull/legs—are not the best option at this stage of training, can be used if it fits your schedule well.

As an intermediate, you can expect to progress every few workouts or weekly. Use the simple intra- and inter-exercise progressions. If these don't work out for you, try to learn how to implement accumulation and intensification phases of training or light/heavy-type work. Of these two, light/heavy implementation is recommended first.

Quality of work is more important than *quantity*. More is not always better, especially in the case of bodyweight work where a significant amount of energy must be spent not only to learn skills correctly but also to perform them correctly. Form deteriorates much more easily with bodyweight work than barbells. If you are thinking about adding more exercises, first consider how your body is reacting.

- Are you making progress week to week?
- How do you feel during the 24-48 hours following a workout?
- Is the quality of other lifestyle factors like sleep, work, and family deteriorating?

It is not a good idea to add exercises if you are losing sleep or are under a high amount of stress, or if you are struggling with soreness of any kind. If a joint is sore, this is especially true. If you are making good progress, why change what is working for you? Stick with what works. It does no good to undertrain, but overtraining can be much more frustrating. However, it is a good idea to push your limits every once in awhile. This will let you see where you are in your training and give you an idea of what you are capable of, but it is very important to back down after pushing yourself past your limits so your body will have proper time to recover. When in doubt, take an extra rest day and see how you feel.

The needs of a person at the *intermediate level* are not quite as simple as the needs of a beginner. At the intermediate level, your training needs begin to diversify based on your specific goals. Therefore, a general routine (such as a full-body routine) will no longer be effective. Your training must become more specialized to fit your goals. This includes not just the strength, hypertrophy, or endurance portions of your workouts but also skill work, sport-specific skills, flexibility, mobility, prehabilitation, and rehabilitation. Here are some examples:

- If your goal is endurance, you should work strength at lower volumes in order to keep efficiency of exercises high and work your specific endurance exercises.
- If your goal is hypertrophy, you should switch from a full-body routine to various splits in most cases. This was discussed specifically in Chapter 5.
- If your goal is strength, you should increase your frequency as much as possible without overtraining or developing overuse injuries.
- You must adapt your frequency, volume, and intensity of training to maintain progress at this ability level. You will also need to learn how to use more complex programming methods.

INTERMEDIATE LEVEL: ENDING A CYCLE

Progression at this stage should generally happen every other workout to weekly. At minimum, you should progress in either repetitions or progressions every couple of workouts to every week or two. Do not be surprised if you do not see the results you expect, especially when you get to levels eight and nine. If you can keep progressing for eight weeks or more and are pain-free, keep doing what you are doing.

If you make absolutely no progress after four weeks, terminate the mesocycle. Remember, fatigue tends to mask fitness. Proper and adequate stimulus must be applied over a long period of time. If you prematurely terminate workouts (at the two or three-week mark), your body might not have received enough accumulated stimuli to adapt. The four-week mark is a good place to terminate, even if no progress is made. You are likely to come back stronger after your recovery week, even if you saw no progress during your cycle. If this is the case, do not consider it a failed cycle.

If, on the other hand, you discover that you have made no progress whatsoever, you should apply additional volume or intensity during the next cycle in order to force adaptations to occur. This can be in the form of adding an additional workout day, adding additional exercises, increasing the number of sets, manipulating exercises with a weight vest to increase intensity, or in another manner.

If you make no progress for two mesocycles in a row, consult the advanced section of this book for options.

Also terminate a mesocycle if your joints or connective tissues begin to feel sore, as this could be a sign of overtraining. These areas tend to be more of a limiting factor than actual nervous system fatigue or muscle recovery during the intermediate stage. A rest week or two may be required to allow for prehabilitation work aimed at healing your joints and connective tissues before you begin a new cycle. It can be difficult for someone who has been working out consistently to back down and rest. You don't need to quit working out altogether; simply remove the offending exercises from your routine and replace them with lower-level progressions and/or prehabilitation work until you can work back up to where you were previously.

ADVANCED LEVEL: EXERCISES AND RECOMMENDATIONS

The advanced level (levels ten through thirteen) is where you will learn the iron cross, planche, and manna. Also, you will develop many combinations on rings/parallettes/floor, from straight body press to handstands. Unless you have insanely good genetics or a great sense for intuitive training, consistency is likely not enough to go far at this point. Your training will now require advanced concepts like periodization and its derivatives (discussed in previous chapters).

The core concepts from lower ability levels like exercise selection and frequency are similar, but you may want to manipulate the volume and intensity by changing around the sets and repetition structure on various days. Variation must be coupled with consistency to produce good results.

The three types of exercises still apply: concentric, isometric, and eccentric. Continue to use the familiar principles outlined in this book. If a technique from another training method has proven successful for you, by all means, incorporate it. *Overcoming Gravity* outlines methods that were successfully used to train the author, his athletes, and first edition users to the advanced level and beyond, but these methods are not the only methods that work. This book's goal is to teach you to think critically, set goals, and build programs that will help you progress toward your own goals, not to proselytize.

By now you will typically be using some form of periodization such as accumulation and intensification or light/heavy routines. If or when you start utilizing more complex periodization methods such as DUP, constructing a routine and then planning a repetition scheme according to type of days takes a little getting used to, particularly if you are accustomed to the failure method. However, you only need to calculate it

once in order to grasp how to do it correctly. Let's say you want ten repetitions for an isometric hold. Your formula will be $10 * 2s = 20s \text{ hold}$. So you will want to pick an isometric hold that you can do for twenty seconds, according to your maximum hold times.

If you select your time in this manner, it will be easy to modify an exercise. To make the exercise more challenging, add ankle weights or a weighted vest; to make it easier, use an assistance method like bands, a pulley system, or assistance from another person. You may have to select straddle planche or straddle front lever instead of the full variation. As you move up toward a high-intensity day (such as a 5x3 block, where you will be performing three repetitions of $3 * 2s = 6s \text{ holds}$), pick the appropriate planche or front lever progression where you can perform all of the holds for six seconds.

Hybrid sets of an exercise can be particularly effective for at the advanced ability level. Hybrid sets are a good solution for isometric exercises that need to be made more challenging. Let's say you were working up to a full planche but could not hold it for the adequate 6-8s needed to advance to the next level. You would begin your set with a maximum hold of 2-3s of the full planche, and then as you fatigue, quickly open up your legs and move into a straddle planche position. Hold this for another period of time to gain additional volume.

These type of exercises work best for advanced athletes; they are not efficient while your technique is still poor. If you have been performing bodyweight exercise with good technique for years, your ability to maintain proper tension will be very useful to correctly implement these techniques. It requires a lot of precision to change from a straddle planche to an advanced tuck planche while maintaining good body position. Beginner and intermediate athletes would usually lack the shoulder and core stabilization required.

Unless the exercise itself is your goal, isolation exercises are only recommended to target weak links in your routine. Most advanced athletes already know their weak links, or at least have inklings of what the weak links in their bodies may be, based on their abilities. For example, you can tell your lats are much more developed than your pecs and biceps for pulling movements if you can feel them activating and contracting more while performing pulling movements! In that case, they will also have greater hypertrophy and strength than your other muscle groups. You can utilize biceps curls as isolation work to specifically amend that biceps weakness. This a good reason to use an isolation exercise.

Here are a few more examples, if you are a gymnast who performs a lot of straight-arm work, then train one-arm pull-ups, your back might be much stronger than your arms. Biceps curls or other biceps exercise may be useful to improve your overall strength. The same is true for barbell lifts like the deadlift, which uses the legs, hips, and back extensively. Many people have a weak link in their back or even their quads if they are posterior chain dominant. Isolation exercises can help keep your body's specific strengths in balance.

Another good reason to use isolations exercises is to lowering the stress level from the compound movements on your connective tissues and targeting the weak links that may be prone to injury. For example, if your elbows get noticeably sore during back lever work, eliminate that work for a period of time and add biceps curls to your workouts instead. In this type of situation, isolation exercises are extremely effective. They should be used sparingly, however, as performing compound exercises still carry the most benefits. Of course, if you are aiming for hypertrophy, isolation work on top of your compound exercises will carry you forward to your goals.

ADVANCED LEVEL: WEEKLY SCHEDULING

It will typically require two to four years of consistent training to reach the advanced level, depending on your genetics, dedication, schedule, and recovery factors like diet and sleep quality. On average, it takes two and a half to four years to reach this level if an athlete is training consistently. Athletes with a higher body mass (175 pounds or more) may take longer—three to five years or longer—to reach this level.

As stated in the intermediate weekly scheduling section of this chapter, adding one extra workout day for every year you have spent working out is a good rule of thumb. If you started out with three workouts per week and have already added a fourth, I would wait two and a half years before adding in a fifth workout per week. Remember the factors we discussed regarding progress and recovery? If you are not getting enough sleep, quality food, or have high amounts of stress, it may not be a good idea to add an additional workout.

If you are training five days a week, the M/Tue/W/F/Sat or Tue/W/Thur/Sat/Sun schedules are good ones. They are basically 3/1/2/1 schedules with the “1” being a rest day. You can also use a 5/2 schedule (working out Monday-Friday with the weekend off) but this is a slightly less effective schedule.

Training more than five days per week is not recommended. It is easy to burn out with a frequency this high. If you want to start playing around with high-frequency routines like two-a-days, please be careful. Utilize shorter cycles (two to four weeks instead of the standard four to eight weeks) and/or significantly reduce the volume per workout.

If you are at the advanced level of strength or beyond and are determined to experiment with six or more workouts per week, utilize one of the five-day schedules first, such as the 3/1/2/1 schedule, with workout days on M/Tue/W/F/Sat. After you have adapted to this schedule, double up on Saturday so that you have morning and evening workouts. Split the volume of one workout into two separate workouts on Saturday to start. You can then ease into doing two full workouts on Saturday. Be very careful when experimenting in this manner! If you push it too much because you want to get stronger more quickly, you could end up doing serious damage to your body. Remember, you can progress with a simple schedule of three workouts per week—up to five if you really want to do more. It puts enormous strain on your body to not have at least two rest days per week. Working out six days a week is tremendously taxing: two-a-days are preferred if you experiment with more frequent workouts.

Use splits if full-body volume is too much. They may be useful at this point in order to allow enough rest time for your body between repeating the same exercises if they are intense enough.

Moving into the *advanced level* brings with it a new set of issues:

- Your training will become even more specific or geared toward your sport.
- Shoring up weak links becomes vastly more important in order to progress. Specific isolation work may be effective in this instance to keep your strengths in balance.
- While sleep, nutrition, and eliminating stress are important for beginners and intermediates, you must get even more serious about these factors at the advanced level. Even a 1% improvement will make an impact at the advanced level, as improving strength and muscle mass is more difficult.
- Keeping a training log is vital at this level, as understanding how your body responds to training is critical. It is important to practice this good habit of logging your training no matter your level. It will allow you to look back and see how your body responds to rests, deloads, exercise intensity, and volume. This will make planning workouts easier.

ADVANCED LEVEL: ENDING A CYCLE

Progression at the advanced level should happen weekly. Add repetitions or increase progression difficulty at least every two weeks. If you are not able to do this after two weeks, some factors of your training are off. Re-examine your use of periodization concepts and observe the results.

As you reach the higher advanced ability levels, you may not notice significant progress until the middle of a mesocycle, the end of a mesocycle, or even after the rest week, unless you are correctly using periodization concepts like DUP. Do not be surprised by this; by now you should have learned patience when it comes to progressing. The strength gains required to jump from one advanced level to the next take a fair amount of time to acquire.

Indications for terminating a cycle at the advanced level are similar to other ability levels. If you make absolutely no progress after four weeks, terminate the cycle. Make sure you wait four weeks, because if you terminate a cycle prematurely, you may not have applied enough of an accumulated stimulus to force your body to adapt. You may come back stronger after the recovery week, even if you did not progress during the cycle. Just as you did in the intermediate stage, if you terminate a cycle, add additional volume or intensity to force adaptations in the next cycle. If you do not make visible progress for two cycles, try a combination of light/heavy exercises with high/low repetitions, respectively. Or, take a close look at your sleep quality, nutrition, and/or stress levels.

Alternatively, you can adjust the programming of your periodization instead of terminating a cycle. Part of your plan was ineffective; if you can refer to your log and determine what adjustment did not work, you can make additional adjustments mid-cycle to get yourself back on track. In this instance, you may want to seek advice from someone more experienced.

Always terminate a mesocycle if your joints or connective tissues get particularly sore, as this may be a sign of overuse rather than simple fatigue or muscle recovery. You may need a week or two of prehabilitation work aimed at healing your connective tissues and joints before you begin another cycle.

INTER-MESOCYCLE FACTORS

There are three main inter-mesocycle factors: *deloading*, *maximal strength testing*, and *workout restructuring*. None of these are specific to beginner, intermediate, or advanced ability levels.

DELOADING

Deloading is an art. The goal is to supercompensate without losing the adaptations you have gained from the previous mesocycle by increasing the amount of rest you are allowing yourself. Unless you have a great coach, it takes a lot of experience to deload well. You need to vary the general protocols to adapt to different situations, depending on recovery factors like time availability, sleep quality, nutrition, etc. Overall, it is more productive to perform some type of exercise during a recovery week than to give yourself complete and total rest. Here are some options that tend to work well:

- **Decreasing the Frequency**—For example: eliminate two days of workouts in your week. If you are on a Monday/Tuesday/Recovery/Friday schedule, this leads to performing only twice during a recovery week, such as Monday/Recovery.
- **Decreasing the Volume**—For example: keep the same workout schedule but eliminate half of the exercises. For example, if you have a routine composed of half isometrics and half full range of motion exercises, eliminate the isometrics. Do this for a week, as it is more productive in most cases to continue working on full range of motion during rest weeks. Continue with your skill work. If a cycle was particularly hard on your body, it may be beneficial to eliminate all of the isometric exercises and focus exclusively on skill work and prehabilitative exercises. This type of deloading would focus on handstands and ring support work, as well as improving shoulder, wrist, back, hip and ankle health and mobility.
- **Decreasing the Intensity**—For example: drop all of the exercise progressions back one level. This has the added bonus of being prehabilitative work, as you are performing an easier exercise for more repetitions during your rest week. This helps deload your joints and connective tissues from the intensity of the exercise, as well as train your joints and connective tissues with a lighter exercise to help injury-proof them.
- **Other Deloading Variations**—Manipulate one of the various factors such as decreasing the frequency, volume, or intensity in various other ways. For example, decreasing the volume has many variations: you can decrease the sets, eliminate exercises, modify the number of repetitions, increase rest times, modify the tempo to be easier, and much more. You can even deload while modifying several of these factors at once. Deloading is not rigid.
- **Rest Weeks**—You can implement more prehabilitative work and stretching protocols to reduce the amount of scar tissue/adhesions in your muscles and get your mobility ready for the next training cycle.
- **Light Exercises**—Perform an activity that you don't normally, such as pick-up basketball or a light run. Make sure you can comfortably carry on a conversation while exercising, in order to prevent it from becoming too intense.
- **Most Common Practice for Deloading**—Train only one or two days during your recovery week and only do one exercise from each of the push, pull, and legs categories. Perform one or two sets not-to-failure and implement the last set to failure. This prevents strength decrements, but maintains a low volume so you can recover well. The rest of the week is best spent on soft tissue massage, mobility work, prehabilitation, and any necessary rehabilitation work. The purpose of rest weeks is for your body to recover and for you to work on any weaknesses that get neglected during full-on training.

MAXIMAL STRENGTH TESTING

Maximal strength testing is useful to find new maxes with exercises after a deload week as your body may supercompensate and gain strength and hypertrophy when you cut down on the training and increase the rest.

This is important because you want to start off your new mesocycle in accordance with your new abilities so as not to undertrain.

Maximal strength testing should be done near the end of a deload week with one rest day before the next cycle begins. For example, if you ended your prior mesocycle on a Sunday and are going to start your new mesocycle on Monday then maximal strength testing should be done that Saturday.

Maximal strength testing for concentric, isometric, and eccentric exercises will give you a general idea of how much you have improved after most of the supercompensation from the previous cycle takes place, allowing the fatigue to dissipate and the fitness to manifest itself. Testing after supercompensation gives you quality numbers on which to base your progressions and hold times for the next cycle.

To perform maximal strength testing, begin with a typical warm-up to make sure you are ready to train. Warm up with five to eight repetitions of a low progression of the skill or strength progression you are testing. Next, perform a few repetitions or short holds of the progression that you are maximally testing. Finally, after three to five minutes of rest, maximally test the exercise or isometric in question. Organize your exercises according to the methods outlined in the strength section on concentric, isometric, and eccentric exercises or whatever methods you are using for your next routine and mesocycle.

The process is relatively simple. In advanced training, it is typical not to see any strength gains until after the supercompensation period takes place.

WORKOUT RESTRUCTURING

Working restructuring occurs when you achieve your goals or have re-evaluated to add or subtract certain exercises. There is no hard or fast rule for setting new goals, except to keep in mind that the key to building strength is repeating movements in a progressive manner. If you have workout attention deficit disorder and frequently switch up your routines, you might build some low- or intermediate-level muscle and strength, but you will find high-level strength elusive.

When you achieve a workout goal (for example, full back lever), you have two options. You can keep it in your workout once a week to maintain it, or you can move on to more progressive goals like front lever and other pulling exercises. This is the great thing about categorizing exercises into a push, pull, and legs system; there is typically some overlap among the different goals in each category. For example, becoming strong in planche pushups will confer strength benefits to your handstand pushups and dips. This means that even if you drop achieved goals from your routine, you will likely be able to perform them later without any practice.

If you would prefer to keep an exercise you have mastered in your routine, perform a low volume of the exercise after your warm-up and before your strength exercises. Typically, if you were performing the full back lever for a total volume of thirty seconds, you would want to add in a couple five to ten-second holds after your warm-up in order to maintain it. This should not detract from your strength workouts since you can already perform the exercise, it will simply maintain your ability to perform it without taking up a full exercise slot in your routine.

Do not change goals every mesocycle. If you have multiple long-term goals, stick with them for at least two or three mesocycles to see significant improvement. If you are unable to reach a goal by then, put it on

hold. Work on something else and come back to it. Putting goals on hold can be particularly effective if they have components beyond strength, such as straight-arm press handstands. These require good flexibility and compression, which you might not yet have, so put them on hold (even though you have the muscle strength) while you develop proper flexibility and compression.

If you feel strongly that you need to go in a different direction with your exercises, evaluate and substitute the exercise or exercises in or out of your routine, but always remember that *being consistent is the best way to make progress*.

Regarding A/B routines (or two different workouts with different exercises), push/pull splits, and upper/lower splits: these are not recommended because you do not get the same frequency with the skills as you would with a full-body routine for beginners. If you crave variety they are certainly options, but you should know that your gains are likely to decrease.

ELITE STRENGTH PROGRAMMING

Elite strength programming depends on the individual more than it does on the programming. This should not deter you from aiming toward this level. For example, some women might be unable to reach high-level rings strength skills like the inverted cross even with decades of perfect training. That is not to say that no woman will ever achieve these strength moves—after all, the competitors in women's Olympic weightlifting are putting up nearly 2x bodyweight snatches (weight to overhead in one movement) and 2.5x bodyweight clean and jerks (weight to overhead in two movements). According to John Gill's website on historical strength performances, Lillian Leitzel was said to have performed twenty-seven dynamic one-arm chin-ups in 1918. Some analysis indicates that this strength feat was equivalent to approximately six regular one-arm chin-ups. These types of strength feats are incredible and elite. Who knows the true capabilities that women have? It is likely that Lillian Leitzel had enough strength for the iron cross!

We must, however, discuss genetic limitations. Some athletes will require ten years or more to reach an elite strength level, whereas those with crazy good genetics can reach this level in two years or less. Coaches have seen athletes walk into the gymnastics gym with very little formal weight or gymnastics training and be able to perform an iron cross or front lever even with no connective tissue strengthening. During one gymnastics summer camp, I once saw an 8-year-old kid perform a round-off back handspring without any spotting because he “felt like it and thought he could do it.” He performed it like he had been training in gymnastics for five years. There are some incredibly talented athletes out there.

If you are one of the athletes who may take a solid decade of training to perform high level strength movements, be aware that while you still have potential for greater strength gains, you are likely to plateau more quickly. This is simply the luck of the draw. Utilize methods like DUP and/or heavy/medium/light days as much as you can, then start looking into more specialized work via either the conjugate or concurrent method of periodization. Supplemental isolation or specific strength work may be necessary as well.

If anyone is at this level with at least this much time spent on training, they should be smart enough to eke out every bit from their nutrition and sleep quality as well. This is a great place to optimize in order to continue to increase your strength.

Perhaps you are progressing up to or beyond this point in a few years without any type of periodization or formal training. If so, be glad you have good genetics. DUP and the programming mentioned above will help you get started once you stall out on whatever training plan you are using.

The FIG COP contains an enormous number of strength moves on rings that are not included in the skill and strength progression charts. Moves such as the maltese, victorian, and some strength combinations you will only see in the Olympics are obtainable if you are willing to put in the time and dedication. If you are at this point, please take someone else under your wing and teach them, as this is how knowledge is most effectively passed on.

A Note from Steven Low:

Based on those I have coached, I am quite confident that everyone can obtain at least level eight to nine on the strength and skill progression charts regardless of body type or physiology. This is true if you are willing to make the commitment to consistent training. The planche may remain elusive, but I am confident that everyone, with proper training, nutrition, and sleep, can accomplish the full back lever, full front lever, straddle planche, and possibly one-arm chin-ups and the iron cross.

Obviously, those with a heavier bodyweight are at a disadvantage, but it is certainly doable. Bert Assirati, one of the famed strongmen of the early 1900's, at 266 pounds, was able to obtain three one-arm chin-ups and the iron cross. He clearly had good genetics on his side. Notwithstanding this, with dedicated training, 99% of people who weigh significantly less than him should be able to attain the same skills. This is why I "never say never" to athletes who weigh in at 200 pounds performing one-arm chin-ups and the iron cross. It may take a while, but it is certainly possible. Likewise, John Gill, the father of modern bouldering (a subtype of rock climbing) was a rings aficionado. He was 6'2" and weighed 185 pounds (183 centimeters and 84 kilograms), but was able to obtain the maltese cross, inverted cross, iron cross, one-arm chin-ups, one-arm front-lever, and other remarkable strength feats.

In the end, my conclusion is this: Train hard and consistently. Deload properly. Get your nutrition, sleep, and stress level under control. If you have extra fat hanging off of you, lose weight so you can perform your exercises more easily. Do not use genetics as an excuse to not have fun and go after your goals. Most people will never even know if their genetics are "good" or not because they fail to commit to their training for more than a couple years. Do not worry about things you cannot control.

One final thing: It must be noted that, for high-end advanced and elite strength athletes, you should follow whatever type of workout you plan. This means that any heavy and intense days must be heavy and intense, and any light or low-volume days need to be kept light and low volume. Since you need to put a high amount of stress on your body to keep adaptations coming, there is less margin for error in programming the higher your level. Likewise, recovery factors must be factored in: nutrition, sleep, and stress. This is not to say you should ignore these at the beginner and intermediate levels; you should have them keyed in at all times for maximal progress. This is why there is the saying: *train hard, rest harder*. You must not forget that in order for adaptations to become visible, you must take a break from your training and rest. Recovery reveals adaptations. A good training and recovery regimen is necessary for progress.

One of the better lessons is taken from the Bulgarian system of Olympic weightlifting. They would train multiple times a day at a high intensity. However, their training was interspersed with pure recovery

factors—mobility work, eating, napping, sleeping. This can be distilled into the saying: *Train. Eat. Sleep. Repeat.* The core concepts that underlie training are universal, and if you try to sidestep them you will fail to improve.

ADDITIONAL PROGRAMMING AND MESOCYCLE CONSIDERATIONS

Structuring a routine does not need to encompass everything. Let's say, for example, that you have many different goals in regards to upper-body pushing. You want to obtain a bodyweight press, 1.5x bodyweight bench press, weighted dip with an extra 25% bodyweight, planche, and a freestanding handstand pushup.

There are only so many goals that you can work at one time without spreading yourself too thin. A good system is two pushing, two pulling, and two legs goals and subsequently structuring a routine toward that maximum by selecting exercises for each of those goals. If you are better conditioned, perhaps you will do seven to nine exercises to work toward those goals, rather than the traditional six. This routine would be performed at a frequency of 3x per week, give or take a workout depending upon the user's conditioning base, other sporting activities, and other factors.

Take this real-world example from about five years ago. Steven Low worked handstand pushups and planche work for six months in a row with proper rest breaks every six to eight weeks. Of course, he had other types of pressing goals, but those two exercises were his main focus for strength at the time. When he went to test other exercises he found that the second time he tried weighted dips he was able to work himself up to ninety pounds for five repetitions. Later, when he tried pressing for the first time after completing more strength work, he was able to perform a bodyweight overhead press easily.

Many of the various forms of pressing-type exercises overlap, sometimes fairly significantly. Therefore, you absolutely do not have to attempt to fit in or work everything at once. Doing so will hinder your training.

Keep things simple. Pick out a small subset of goals that you want to work toward. Work those goals religiously for a whole cycle. You should see good, consistent progress toward those goals. If, at the end of the cycle, you want to change your goals and change your exercises, feel free to do so. If you want to strength test other movements, that also works. You do not have to strength test only the movements you were working on during the cycle.

Adjustments can be made during the deload week if you want to change your focus. That is also the time to focus on different aspects of your development if you are considering changing them. In terms of programming, here is a generalized model that can be followed:

1. Linear progression or linear repetition progression is first. If possible, add repetitions or improve your progressions every workout. Do this for most of the beginner phase of training.
2. Simple intra- and inter-exercise progressions are next. You will want to increase the number of repetitions or progressions every other workout or so. Do this from the end of the beginner phase of training through the beginning of the intermediate phase.
3. Simple periodization is next. Utilize accumulation and intensification phases in training or the light/heavy model of changing the repetitions from workout to workout. Do this during the mid-intermediate to late-intermediate phase of training.

4. Daily undulating periodization (DUP) works exceedingly well as you move into the upper intermediate and advanced phases of training.
5. When you are midway through the advanced level of training or beyond, a hybridized light/heavy DUP type system is effective. You may also want to start experimenting with concurrent or conjugate models of periodization if you are having trouble progressing.
6. At the elite level of training, a concurrent or conjugate system will likely work best.

In conclusion, do not make things too complicated. Selecting a few goals (instead of many) and building a routine around them makes your workout routine simple and easy to accomplish. The level of complexity in programming should only be advanced when absolutely necessary; you do not need these complexities to make progress.

Once you have thought “Maybe I should do” or “Maybe it would be better if I changed this...” three or four times, you are probably over-thinking it. It is true that it takes a lot of practice to succeed, but over-thinking your programming will cause you to focus too much of your time on making your routine “perfect.” The biggest factor for progress is consistency, not having a perfect routine.

Another reason to keep things simple is that introducing too many factors at once hurts your ability to analyze your routines and modify them for your next cycle. If you only change a few variables in your next cycle, you can more accurately pinpoint your level of improvement and which changes were major contributors to that improvement. If you change multiple variables at once, how will you know what actually helped you improve? Even more complicated is the scenario that if you made absolutely no progress, what were the variables that were helping and which were hindering you?

Keep whatever you do as simple as possible. Follow the KISS model—*keep it simple, stupid*. Make very few changes to your overall programming from cycle to cycle so you can see what affects your progress. This allows you to become a better programmer much faster, and it will also help you overall in making faster progress.

STOP READING. TAKE ACTION.

Now that you have planned the core portion of your workouts, you can begin to determine which factors need to be evaluated during and after a mesocycle.

Bookmark this page or make note of it in your workout log. You should refer back to this section (part two—chapters seven through twelve) to re-evaluate your goals, exercise, and progress. Do this weekly if you are a beginner or intermediate athlete, or bi-weekly to monthly if you are advanced.

The material contained in part two of *Overcoming Gravity* is dense, making certain portions difficult to comprehend the first time you read it. If your schedule allows, re-read any chapters that you had difficulty understanding. Knowing and applying these concepts will help you improve your workouts so that you can progress more effectively.

CHAPTER 12 SUMMARY

MESOCYCLE PLANNING

In this chapter, we looked at the different levels of strength programming both during and after a mesocycle. We noted how the weekly scheduling of workouts via frequency and overall volume via amounts of exercises, intensity, repetitions, and sets all affect how progress is made at these different levels. We observed that beginners tend to progress the fastest and require less advanced programming, while intermediate and advanced athletes may require more progressive loading or complexity in programming.

We discussed the different factors related to exercise choice and when to terminate cycles.

We discussed deloading, maximal strength testing, workout restructuring, elite strength and how to apply these concepts to your rest week and how to integrate them into the mesocycle that follows. Deloading allows supercompensations to occur and affords a rest break for additional maximal strength testing. Workout restructuring can be utilized if you want to focus on different goals or implement different exercises to work toward the same goals. Finally, it takes a great deal of time to reach the elite strength ability level due to various inherent and environmental factors.

